Task 4: Banking System

CodeAlpha - C++ Programming Internship

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Objective

A console-based banking application in C++ that lets a user open accounts, view account details, deposit and withdraw funds, and search for an account by its number.

Features

Menu option	Function	What it does
1. Open an account	openAccount()	Takes account number, holder name and initial balance; stores it in a vector
2. Show account details	searchAccount() / displayAccount()	Prints number, holder name and balance for a given account
3. Deposit Funds	deposit()	Adds the amount to the matching account balance
4. Withdraw Funds	withdraw()	Subtracts the amount if balance is sufficient, else shows an error
5. Search for an account	searchAccount()	Finds an account by number or reports "Account not found!"
6. Exit	-	Ends the program

Concepts used

struct, std::vector, functions with pass-by-reference, range-based for loop, do-while loop, switch-case, and console input/output (cin, getline, cout).

Source Code (bank_app.cpp)

```
#include <iostream>
#include <string>
#include <vector>

using namespace std;

struct BankAccount {
    int accountNumber;
    string accountHolderName;
    double balance;
};

void openAccount(vector<BankAccount>& accounts) {
    BankAccount newAccount;
    cout << "Enter account number: ";
    cin >> newAccount.accountNumber;
    cin.ignore();
    cout << "Enter account holder name: ";
    getline(cin, newAccount.accountHolderName);
    cout << "Enter initial balance: ";
    cin >> newAccount.balance;
    accounts.push_back(newAccount);
    cout << "Account created successfully!" << endl;
}

void displayAccount(const BankAccount& account) {
    cout << "Account Number: " << account.accountNumber << endl;
    cout << "Account Holder Name: " << account.accountHolderName << endl;
    cout << "Balance: " << account.balance << endl;
}

void deposit(vector<BankAccount>& accounts, int accountNumber, double amount) {
    for (BankAccount& account : accounts) {
        if (account.accountNumber == accountNumber) {
            account.balance += amount;
            cout << "Deposit successful!" << endl;
            return;
        }
    }
    cout << "Account not found!" << endl;
}

void withdraw(vector<BankAccount>& accounts, int accountNumber, double amount) {
    for (BankAccount& account : accounts) {
        if (account.accountNumber == accountNumber) {
            if (account.balance >= amount) {
                account.balance -= amount;
                cout << "Withdrawal successful!" << endl;
            } else {
                cout << "Insufficient Balance!" << endl;
            }
            return;
        }
    }
    cout << "Account not found!" << endl;
}

void searchAccount(const vector<BankAccount>& accounts, int accountNumber) {
    for (const BankAccount& account : accounts) {
        if (account.accountNumber == accountNumber) {
            displayAccount(account);
            return;
        }
    }
    cout << "Account not found!" << endl;
}
int main() {
```

```
    vector<BankAccount> accounts;
    int choice;

    do {
        cout << "\n***** BANKING APP *****" << endl;
        cout << "1. Open an account" << endl;
        cout << "2. Show account details" << endl;
        cout << "3. Deposit Funds" << endl;
        cout << "4. Withdraw Funds" << endl;
        cout << "5. Search for an account" << endl;
        cout << "6. Exit" << endl;
        cout << "Enter Your choice: ";
        cin >> choice;

        switch (choice) {
            case 1: {
                openAccount(accounts);
                break;
            }
            case 2: {
                int accountNumber;
                cout << "Enter account Number: ";
                cin >> accountNumber;
                searchAccount(accounts, accountNumber);
                break;
            }
            case 3: {
                int accountNumber;
                double amount;
                cout << "Enter account Number: ";
                cin >> accountNumber;
                cout << "Enter the amount to deposit: ";
                cin >> amount;
                deposit(accounts, accountNumber, amount);
                break;
            }
            case 4: {
                int accountNumber;
                double amount;
                cout << "Enter account Number: ";
                cin >> accountNumber;
                cout << "Enter the amount to withdraw: ";
                cin >> amount;
                withdraw(accounts, accountNumber, amount);
                break;
            }
            case 5: {
                int accountNumber;
                cout << "Enter account Number: ";
                cin >> accountNumber;
                searchAccount(accounts, accountNumber);
                break;
            }
            case 6: {
                cout << "Exiting the Program..." << endl;
                break;
            }
            default:
                cout << "Invalid choice! Please try again." << endl;
        }
    } while (choice != 6);

    return 0;
}
```

Sample Output

Test run: open account 123, deposit 500, withdraw 200, try to withdraw 5000, show details, search for a missing account, exit.